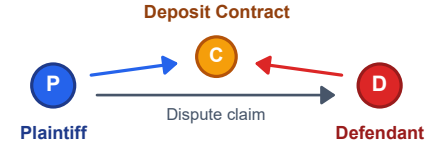
Stage 1: Dynamic Legal Graph Extraction

Court Record $\rightarrow \mathcal{G}(t) = (\mathcal{V}, \mathcal{E}, \mathcal{T})$

Bipartite Dispute Subgraph



Chronological Event Stream

- t_0: **Contract Execution** — Valid consent, Art. 117
- t_1: **Default Event** — Non-performance, Art. 328
- t_2: **COVID-19 Freeze** — Tolling $\frac{d\tau}{dt} = 0$, Art. 156
- t_3: **Lawsuit Filing** — Filing to Court $\Delta\tau$

$\{(u, v, t)\}$

Stage 2: Axiom-TGAT Attention

Continuous Harmonic Deontic Attention

Bochner Spectral Time Encoder

$$\Phi(\Delta\tau) = \sqrt{\tfrac{2}{d_T}}[\cos(\omega_k \Delta\tau) \,\|\, \sin(\omega_k \Delta\tau)]^T$$

Continuous Fourier decomposition via Bochner's Theorem

Dual-Clock Mapping: $\Delta\tau = \tau(t_{\text{file}}) - \tau(t_{\text{default}})$

Temporal Cross-Attention

$$\alpha_{uv}(t) = \text{Softmax}\left(\frac{\mathbf{q}_u^T[\mathbf{k}_v \,\|\, \Phi(\Delta\tau)]}{\sqrt{d_k}}\right)$$

$$\mathbf{h}_u(t) = \sum_{v \in \mathcal{N}(u)} \alpha_{uv}(t)(\mathbf{W}_V[\mathbf{z}_v(t) \,\|\, \Phi(\Delta\tau)])$$

Inductive 2-hop temporal neighborhood aggregation

Dynamic Node Representation $\mathbf{z}_{uv}(t)$

$$\mathbf{z}_{uv}(t) = \text{MLP}([\mathbf{h}_u(t) \,\|\, \mathbf{h}_v(t) \,\|\, \mathbf{e}_{uv}(t)])$$

$\mathbf{z}_{uv}(t)$

Stage 3: Ontological Regularization

OWL 2 DL TBox Loss & Normative Arbiter

Vietnamese Civil Law Ontology (VCLO)

$\mathcal{T}_1$: Dispute $\subseteq \exists$hasDefault. Date

$\mathcal{T}_2$: TimeBarred $\subseteq \neg$EnforceableRelief

$\mathcal{T}_3$: Perf $\geq 2/3 \subseteq$ CuredDefect (Art. 129(2))

Normative Hierarchy DAG Arbiter (NH-DA)

Lex Specialis derogat legi generali

General Civil Code (BLDS 2015) $\succ$ Specialized Law (Land / Credit Law)

Description Logic Loss $\mathcal{L}_{\text{onto}}$

$$\mathcal{L}_{\text{onto}} = \sum_{C \sqsubseteq D} \max(0, P(\neg D \mid \mathbf{z}) - P(\neg C \mid \mathbf{z}))^2$$

Zero tolerance for illegal judicial contradictions

$\mathcal{L}_{\text{onto}}$

Stage 4: Defeasible Adjudication

Phan Minh Dung Abstract Argumentation

Defeasible Argumentation $\mathcal{AF} = (\mathcal{A}rgs, \mathcal{A}tt)$



Knaster-Tarski Grounded Extension

$$\mathcal{GE}(\mathcal{AF}) = \text{lfp}(F_{AF}) = \{A_3\}$$

Procedural bar defeats substantive relief

AUDITED JUDICIAL VERDICT

$\hat{y} \in \{0, 1, 2, 3\}$ **+ Proof Certificate**

90.17% Hits@1 | 99.87% LCR | 0% Violation